NOTEBOOK FOR MA215 PROBABILITY

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Contents

Preface **4**

1 Counting and Probability Spaces **5**
1.1 Counting principles and combinatorial models 5
1.2 Events, probability measures, and set identities 6

2 Conditional Probability and Independence **8**
2.1 Conditional probability and the multiplication rule 8
2.2 Independence and Bayes' theorem 8

3 Random Variables and Probability Distributions **9**
3.1 Discrete variables and probability masses 9
3.2 Densities and continuous variables 10
3.3 Cumulative distribution functions 10
3.4 Standard discrete distributions 12
3.5 Standard continuous distributions 13

4 Expectation, Moments, and Counting Methods **15**
4.1 Expectation, integrability, and variance 15
4.2 Moments of standard distributions 17
4.3 Indicators, waiting times, and applications 17
4.4 Inclusion–exclusion through indicators 19
4.5 Expectations from tail probabilities 20

5 Joint Distributions and Dependence **20**
5.1 Joint laws, marginalization, and independence 20
5.2 Gaussian models and covariance 23
5.3 Variance of sums and correlation 24

6 Transformations and Sums of Random Variables **26**
6.1 One-dimensional transformations and quantiles 26
6.2 Multivariate transformations and simulation 27
6.3 Gaussian moments from polar coordinates 28
6.4 Convolution of independent variables 28
6.5 Closure properties of discrete distributions 29

7 Conditional Distributions and Conditional Expectation **30**
7.1 Conditional mass functions and densities 30
7.2 Conditional Gaussian laws and support constraints 32
7.3 Conditional expectation and recursive models 32
7.4 Decomposing variance by conditioning 34

8 Order Statistics **34**
8.1 Joint and marginal densities of ordered samples 34
8.2 Extremes and the Beta family 35

9 Generating Functions and Characteristic Functions **36**
9.1 Probability generating functions 36
9.2 Moment generating functions 37
9.3 Characteristic functions and inversion 37

10 Probability Inequalities **38**
10.1 Markov and Chebyshev inequalities 38
10.2 Chernoff bounds 39
10.3 Convexity and Jensen's inequality 39

11 Convergence and Limit Theorems **40**
11.1 Modes of convergence 40
11.2 Weak and strong laws of large numbers 41
11.3 The central limit theorem and transform methods 42
11.4 Almost sure events and dominated convergence 43

A Answers and Solution Sketches **45**

Preface

These are personal lecture notes for **Prof. Jieliang Hong's MA215 Probability course** at Southern University of Science and Technology (SUSTech) in the Fall 2024 semester.

Before reading, it is important to note that the lecturer does not endorse these notes and that I have modified them (often significantly) after each lecture. They are nowhere near accurate representations of what was actually lectured, and in particular, **all errors are almost surely mine**.

I would also like to acknowledge the great help of AI (mainly ChatGPT 6-Astra) in generating TikZ code, creating illustrations, and improving the clarity of my writing.

If you find typos or errors in these notes, please get in touch with `12311501@mail.sustech.edu.cn`. Suggestions and corrections are welcome.

How to use these notes

The material is organized by topic rather than by lecture date. Sections 1–3 introduce probability models and distributions; Sections 4–8 develop expectation, dependence, conditioning, and transformations. Sections 9–11 cover transforms, inequalities, and asymptotic behavior. Solution sketches appear in Appendix A.

Reading the proofs. Detailed proofs emphasize reusable methods: disjoint decompositions, indicators, conditioning, and transforms. Shorter proof sketches isolate the main idea. The strong law and the continuity theorem use explicitly stated auxiliary results; their sketches are not self-contained treatments of those advanced tools.

Throughout, $\text{Exp}(\lambda)$ and $\text{Gamma}(\alpha, \lambda)$ use a rate parameter λ, and $\text{Geometric}(p)$ counts trials up to and including the first success. These conventions determine the corresponding moment formulas.

1 Counting and Probability Spaces

A probability model requires both a description of outcomes and a consistent rule for assigning probabilities. Counting supplies the first examples; probability spaces extend the same principles beyond finite experiments.

1.1 Counting principles and combinatorial models

Theorem 1.1. Basic principle of counting
Suppose there are two experiments. Experiment 1 has n possible outcomes, and for each such outcome, experiment 2 has m possible outcomes.
Then, together, there are $m \times n$ possible outcomes.

This fundamental theorem could be extended to many finite experiments by Induction.

Definition 1.1. Permutation
Permutation means the different ordered arrangements of objects.

Theorem 1.2. Suppose we have n objects. Then there are $n! = \prod_{i=1}^{n}(i) = 1 \times 2 \times \cdots \times n$ possible permutations.

Theorem 1.3. There are n objects, of which n_1 are alike, n_2 are alike,...,n_r are alike. Then there are $\frac{n!}{n_1! \times n_2! \times \ldots n_r!}$ possible outcomes.

Proof sketch. Temporarily label all n objects, giving $n!$ permutations. Erasing the labels identifies exactly $n_1! \cdots n_r!$ labelled permutations with each distinct arrangement: labels can be permuted independently within each class. Dividing gives the formula, assuming $n_1 + \cdots + n_r = n$. □

Definition 1.2. Combination
A combination refers to selecting items from a set where order does not matter.

Theorem 1.4. If we choose r objects from a total of n distinct objects at a time, then the number of possible combinations is $\binom{n}{r}$

Theorem 1.5. Binomial Theorem
For any positive integer $n \geq 1$

$$(x+y)^n = \sum_{k=0}^{n}(\binom{n}{k}x^k y^{n-k})$$

Proof sketch. Expand the product of n factors $(x + y)$. A term $x^k y^{n-k}$ arises by choosing x from exactly k factors and y from the rest. There are $\binom{n}{k}$ such choices, which gives its coefficient. □

Definition 1.3. Induction
Mathematical Induction is a proof method for natural numbers, consisting of a base case and an inductive step to show a statement holds for all natural numbers.

Mathematical Induction's basic step:

1. Basic step: The case holds when $n = 1$
2. Inductive step: Assume $n = k$ holds for some $k \geq 1$. Then $n = k + 1$ holds.

Quiz 1.1. A committee of 3 men and 3 women will be formed from 8 women and 6 men. How many different committees?

1. 2 of the men refuse to serve together?
2. 2 of the women refuse to serve together?
3. 1: A man and a woman refuse to serve together?

1.2 Events, probability measures, and set identities

Probability Space includes Sample Space, Events, and Probability Measure.
Probability Space is a special case of measure theory.

Definition 1.4. Sample Space
The sample space S is the set of all possible outcomes of an experiment.

Definition 1.5. Event
An event is a subset of the sample space S, denoted $E \subset S$

Definition 1.6. Set Operation
Let E and F be two events, and S be the sample space.

1. **Union**: $E \cup F = \{x | x \in E \text{ or } x \in F\}$
2. **Intersection**: $E \cap F = \{x | x \in E \text{ and } x \in F\}$
3. **Complement**: $E^c = \{x | x \notin E \text{ and } x \in S\}$
4. **Difference**: $E \setminus F = \{x \mid x \in E \text{ and } x \notin F\}$

Definition 1.7. Extension: $\sigma -$ algebra
Let $\mathcal{X}$ be a non-empty set. $\mathcal{F}$ is said to be a σ-algebra if:

1. $\mathcal{X} \in \mathcal{F}$
2. If $A \in \mathcal{F}, A^c \in \mathcal{F}$
3. If $A_1, A_2 \cdots \in \mathcal{F},$ then $\bigcup_{i=1}^{\infty}(A_i) \in \mathcal{F}$

Theorem 1.6. De Morgan's Law
For each $n \geq 1$, we have

$$(\bigcup_{i=1}^{n}(E_i))^c = \bigcap_{i=1}^{n}(E_i^c)$$

$$(\bigcap_{i=1}^{n}(E_i))^c = \bigcup_{i=1}^{n}(E_i^c)$$

Axiom 1.1. Axiom of Probability Let S be a sample space. For each event E, the probability $P(E)$ satisfies:

1. $0 \leq P(E) \leq 1$
2. $P(S) = 1$
3. For any sequence of mutually exclusive events $E_1, E_2 \ldots$, we have:

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

We called it axiom because it is the axiomatical way of defining the probability measure and it is the base for the probability

Theorem 1.7. Basic corollaries:

1. $P(E) = 1 - P(E^c)$
2. If $E \subset F$, then $P(E) \leq P(F)$
3. $P(E \cup F) = P(E) + P(F) - P(E \cap F)$
4. Inclusion-Exclusion Identity:(Extension of the line above)

$$P(\bigcup_{i=1}^{n} E_i) = \sum_{i=1}^{n} P(E_i) - \sum_{i_1<i_2} P(E_{i_1} \cap E_{i_2}) + \cdots + (-1)^{n+1} P(\bigcap_{i=1}^{n}(E_i))$$

Proof sketch. The disjoint decompositions $S = E \cup E^c$ and $F = E \cup (F \setminus E)$ (when $E \subseteq F$) give the first two claims by additivity and non-negativity. For the third, decompose $E \cup F$ into $E \setminus F$, $E \cap F$, and $F \setminus E$. The general inclusion–exclusion identity follows by expanding a product of indicators; the full argument appears in Theorem 4.3. □

Quiz 1.2. There are N cards numbered as $1, 2, \ldots, N$. Pick 1 card uniformly at random. Write down the number and return the card. Repeat for n times ($n > N, n = N, n < N$), we get a sequence $(x_1, x_2 \ldots, x_n)$.

1. P(the sequence is strictly increasing)
2. P(the sequence is non-decreasing)

2 Conditional Probability and Independence

Conditioning updates a model when information becomes available. Independence identifies the special case in which this update leaves probabilities unchanged.

2.1 Conditional probability and the multiplication rule

Definition 2.1. Conditional Probability
For two events E, F such that $P(E) > 0$. The conditional probability that F occurs given that E has occurred is denoted by:

$$P(F|E) = \frac{P(F \cap E)}{P(E)}$$

This definition gives a new perspective on the conditional probability:

Theorem 2.1. If each outcome of a finite sample space is equally likely, we may compute the conditional probability of the form $P(F|E)$ by using E as the reduced sample space.

Theorem 2.2. Multiplication Law
For events E and F, we have:

$$P(E \cap F) = P(E) \times P(F|E)$$

More generally:

$$P(E_1 \cap E_2 \cap \cdots \cap E_n) = P(E_1) \times P(E_2|E_1) \times P(E_3|(E_1 \cap E_2)) \ldots P(E_n|\bigcap_{i=1}^{n-1}(E_i))$$

Proof. Let $C_k = E_1 \cap \cdots \cap E_k$. Whenever $P(C_{k-1}) > 0$, the definition of conditioning gives

$$P(E_k \mid C_{k-1}) = \frac{P(C_k)}{P(C_{k-1})}.$$

Multiplying these ratios for $k = 2, \ldots, n$ and then multiplying by $P(E_1)$ cancels all intermediate terms, leaving $P(C_n)$. The two-event formula is the case $n = 2$. □

The multiplication identity in Theorem 2.2 is the **chain rule**. Each conditional probability is used only when its conditioning event has positive probability.

2.2 Independence and Bayes' theorem

Definition 2.2. Independence
For two events E and F, independence means the following identity. If $P(E) > 0$, it is equivalent to $P(F \mid E) = P(F)$.

$$P(E \cap F) = P(E)P(F)$$

Theorem 2.3. Total Probability Formula
Let $A_1, A_2, \ldots, A_n$ be mutually exclusive with $S = \bigcup_{k=1}^{n}(A_k)$.
Then $\forall$ event B:

$$P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i)$$

Proof. The events $B \cap A_i$ are disjoint and their union is B. Therefore

$$P(B) = \sum_i P(B \cap A_i) = \sum_i P(B \mid A_i)P(A_i).$$

Cells with $P(A_i) = 0$ contribute zero and are omitted from the conditional-probability expression. □

Theorem 2.4. Bayes's Theorem
Let $A_1, A_2, \ldots A_n$ be mutually exclusive so that $S = \bigcup_{k=1}^{n}(A_k)$.
Assume $P(B) > 0$ and omit partition cells of probability zero. Then:

$$P(A_j|B) = \frac{P(B|A_j) \times P(A_j)}{\sum_{i=1}^{n} P(B|A_i)P(A_i)}$$

Proof. For $P(B) > 0$, write $P(A_j \mid B) = P(A_j \cap B)/P(B)$. The multiplication rule gives the numerator $P(B \mid A_j)P(A_j)$, and total probability gives the denominator. Thus Bayes' formula reverses the direction of conditioning by normalizing the joint probabilities. □

Quiz 2.1. A gambler has a fair coin and a two-headed coin in his pocket.

1. He selects one of the coins randomly; when he flips it, it shows heads. What is the probability that it is a fair coin?
2. Suppose that he flips the same coin a second time and, again, it shows heads. Now, what is the probability that it is a fair coin?
3. Suppose that he flips the same coin a third time, and it shows tails. Now, what is the probability that it is a fair coin?

3 Random Variables and Probability Distributions

Random variables extract numerical information from outcomes. Probability masses, densities, and cumulative distribution functions are complementary descriptions of their laws.

3.1 Discrete variables and probability masses

Definition 3.1. Discrete Random Variable
A random variable is a measurable function $X : S \to \mathbb{R}$. It is discrete if it takes values in a countable set with probability one.

For example, 80 students, of whom 70 are male. Choose one uniformly at random. Repeat independently 4 times, with replacement. Let $X = \#$ of male students chosen.

Then X is a discrete R.V. taking values in $\{0, 1, 2, 3, 4\}$

Moreover:

$$\forall\ k \in \{0, 1, 2, 3, 4\}. P(X = k) = \binom{4}{k} (\frac{7}{8})^k (\frac{1}{8})^{4-k}$$

This is the probability mass function of X.

Definition 3.2. Probability Mass Function
For a discrete random variable X, we can define the probability mass function(p.m.f), where $p(m)$ of X by

$$p(m) = P(X = m)$$

3.2 Densities and continuous variables

Definition 3.3. Probability Density Function
A non-negative function $f : (-\infty, \infty) \longrightarrow [0, \infty]$ is called a probability density function(p.d.f.), if:

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

Definition 3.4. Continuous Random Variable
A random variable X is called a continuous random variable if exists a p.d.f. f such that:

$$\forall a \leq b,\ P(a \leq X \leq b) = \int_a^b f(x)\, dx$$

Remark: Let $b = a$ to get:

$$P(X = a) = P(a \leq X \leq a) = \int_a^a f(x)\, dx = 0$$

3.3 Cumulative distribution functions

Definition 3.5. Cumulative Distribution Function
For a random variable X, the cumulative distribution function(c.d.f.) of X is:

$$F_X(b) = P(X \leq b)$$

We notice that:

1. For discrete random variable:

$$F(b) = \sum_{x \in \mathcal{S}_X,\, x \leq b} P(X = x)$$

2. For continuous random variable:

$$F'(b) = f(b) \quad \text{almost everywhere}$$

However, random variables have forms other than these two kinds. See the quiz below:

Quiz 3.1. The cumulative distribution function of X is

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{x}{2} & 0 \le x < 1 \\ \frac{2}{3} & 1 \le x < 2 \\ \frac{11}{12} & 2 \le x < 3 \\ 1 & x \ge 3 \end{cases}$$

(i) $P(X < 3)$

(ii) $P(X = 1)$

(iii) $P(X > \frac{1}{2})$

Theorem 3.1. 1. If $A_n \subset A_{n+1}, \forall n \ge 1$, then:

$$P(\bigcup_{n=1}^{\infty} A_n) = \lim_{n\to\infty} P(A_n)$$

2. If $B_{n+1} \subset B_n, \forall n \ge 1$, then:

$$P(\bigcap_{n=1}^{\infty} B_n) = \lim_{n\to\infty} P(B_n)$$

Proof. For an increasing sequence, set $D_1 = A_1$ and $D_n = A_n \setminus A_{n-1}$ for $n \ge 2$. These events are disjoint, $A_n = \bigcup_{k=1}^{n} D_k$, and $\bigcup_n A_n = \bigcup_n D_n$. Countable additivity gives

$$P\left(\bigcup_n A_n\right) = \sum_{k=1}^{\infty} P(D_k) = \lim_{n\to\infty} \sum_{k=1}^{n} P(D_k) = \lim_{n\to\infty} P(A_n).$$

For a decreasing sequence B_n, the complements B_n^c increase. Apply the first result to them and use $P(B_n) = 1 - P(B_n^c)$ and De Morgan's law. □

Theorem 3.2. Properties of Cumulative Distribution Function:
Let F be a cumulative distribution function.

1. F is a non-decreasing function, i.e.:

$$\forall\ a < b, F(a) \le F(b)$$

2. $\lim_{b\to-\infty} F(b) = 0, \lim_{b\to\infty} F(b) = 1$

3. F is right continuous, i.e.:

$$b_n \downarrow b \quad \Longrightarrow \quad \lim_{n\to\infty} F(b_n) = F(b)$$

4. F has left limits, i.e.:

$$a_n \uparrow a,\ a_n < a \quad \Longrightarrow \quad \lim_{n\to\infty} F(a_n) = F(a^-) = P(X < a)$$

Proof. If $a < b$, then $\{X \le a\} \subseteq \{X \le b\}$, proving monotonicity. Since X is real-valued, $\{X \le n\} \uparrow S$ and $\{X \le -n\} \downarrow \varnothing$; continuity of probability gives the two limits at infinity.

If $b_n \downarrow b$, then $\{X \le b_n\} \downarrow \{X \le b\}$, which proves right continuity. If $a_n \uparrow a$ with $a_n < a$, then $\{X \le a_n\} \uparrow \{X < a\}$, giving the left limit. In particular,

$$P(X = a) = F(a) - F(a^-).$$

Thus a jump of the c.d.f. records exactly the probability mass at that point. □

These properties follow from continuity of probability along increasing or decreasing sequences of events.

For example, we take $X \sim Bernoulli(p)$, then:

$$F_X(b) = \begin{cases} 1 & b \ge 1 \\ 1-p & 0 \le b < 1 \\ 0, & b < 0 \end{cases}$$

It is a very traditional step function.

3.4 Standard discrete distributions

Definition 3.6. Bernoulli and binomial distributions

1. A random variable is said to be a **Bernoulli** random variable with parameter $p \in [0, 1]$ if:

$$P(X = 0) = 1 - p \text{ and } P(X = 1) = p$$

We say $X \sim \text{Bernoulli}(p)$

2. If we toss a coin independently for n times and let $X = \#$ of heads coming up, then X is said to be a **Binomial** random variable with parameter $p \in [0, 1]$
Denoted by $X \sim \text{Bin}(n, p)$

The probability mass function of $\text{Bin}(n, p)$ is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

For example, Alice is in a class of 80 students after 100 independent trials. We count X as the # of times where Alice is picked. Then $X \sim \text{Bin}(100, \frac{1}{80})$

Remark: A binomial random variable is the sum of n independent Bernoulli random variables with the same success probability p.

Definition 3.7. Poisson Random variable
Let $X = \text{Bin}(n, \frac{\lambda}{n})$ for some $\lambda > 0$.
Then let $n \to \infty$, we can get a new p.m.f, which is the p.m.f of Poisson R.V. :

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!} \ \forall k \geq 0$$

Denoted by $X \sim \text{Poisson}(\lambda)$

Definition 3.8. Geometric Random Variable
There is a coin having probability $p \in (0, 1)$ of coming up heads. Toss the coin until it shows up on the head. Let $X = \#$ of tosses needed.
Then $X \sim \text{Geometric}(p)$, then p.m.f. of which is :

$$P(X = k) = (1 - p)^{k-1}p \ \forall k \geq 1$$

Denoted by $X \sim \text{Geometric}(p)$

We count trials up to and including the first success, so the support is $\{1, 2, \ldots\}$. The alternative convention counts failures before the first success and equals $X - 1$.

Quiz 3.2. Jim is conducting a random walk on the real line starting from 0. Each time, independently of anything else, he moves one step to the right with probability p and to the left with probability $1 - p$. Let X_n be Jim's position at time n. Find $P(X_n = k)$ for each $-n \leq k \leq n$

3.5 Standard continuous distributions

Definition 3.9. Uniform Random Variable
A random variable $X \sim Uniform(\alpha, \beta)$, if the p.d.f. of X is:

$$f(x) = \frac{1}{\beta - \alpha}1_{(\alpha,\beta)}(x) = \begin{cases} \frac{1}{\beta-\alpha} & \text{if } \alpha < x < \beta \\ 0 & \text{, otherwise} \end{cases}$$

Indicator function:

$$1_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

It is also called a characteristic function.
For example: Let $X \sim Unif(1, 5)$, find $P(X > 3.5)$

Solution:

$$P(X > 3.5) = P(X \geq 3.5)$$

$$P(X \geq 3.5) = \lim_{b \to \infty} P(3.5 \leq X \leq b)$$

$$P(X \geq 3.5) = \int_{3.5}^{\infty} f(x)\, dx = \frac{3}{8}$$

Definition 3.10. Exponential Random Variable
We say a X is an exponential random variable with parameter $\lambda > 0$ if the p.d.f. Is:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Definition 3.11. Memoryless
We say a random variable is memoryless if:

$$P(X > t + s | X > t) = P(X > s)\ \forall t, s > 0$$

It is easy to prove that all exponential random variables are memoryless.

Theorem 3.3. If X is non-negative, non-degenerate, continuous and memoryless, then $X \sim Exp(\lambda)$ for some $\lambda > 0$.
The continuity assumption excludes the discrete geometric distribution.

Proof. Let $S(t) = P(X > t)$ for $t \geq 0$. Continuity and non-negativity of X give $S(0) = 1$. Memorylessness gives $S(s + t) = S(s)S(t)$ whenever the conditioning probability is positive. In fact $S(t) > 0$ for every finite t: otherwise choose m large enough that $S(t/m) > 0$ by continuity at zero, and iterate the identity to obtain $S(t) = S(t/m)^m > 0$, a contradiction.

Set $h(t) = -\log S(t)$. Then $h(s + t) = h(s) + h(t)$, and h is continuous. First for non-negative rational t, and then by continuity for all $t \geq 0$, this implies $h(t) = \lambda t$ with $\lambda = h(1)$. Since $S(t) \to 0$ as $t \to \infty$, we have $\lambda > 0$. Hence $S(t) = e^{-\lambda t}$, the exponential survival function. Conversely, this survival function satisfies the memoryless identity directly. □

For example, let $X \sim \text{Exp}(\lambda)$, where $\lambda > 0$ is the rate. Then

$$P(50 \leq X \leq 150) = e^{-50\lambda} - e^{-150\lambda}.$$

If, in addition, $E[X] = 100$, then $\lambda = 1/100$. Normalization alone does not determine the rate.

Definition 3.12. Gamma function:

$$\Gamma(\alpha) = \int_0^{\infty} e^{-y} y^{\alpha - 1}\, dy$$

Moreover:

$$\text{If } \alpha = n \in \mathbb{N}, \text{ then } \Gamma(n) = (n-1)!$$

Definition 3.13. Gamma Random Variable
Let X be a Gamma Random Variable, denoted by $X \sim Gamma(n, \lambda)$, where $n > 0$ is the shape and $\lambda > 0$ is the rate. Its p.d.f. is:

$$f(x) = \frac{\lambda^n x^{n-1} e^{-\lambda x}}{\Gamma(n)}, \quad x > 0$$

In fact, if $X_1, X_2, \ldots, X_n$ are independent $Exp(\lambda)$, then

$$X_1 + X_2 + \cdots + X_n \sim \text{Gamma}(n, \lambda)$$

We can understand the Gamma random variable in both two ways.

Definition 3.14. Normal Random Variable
We say a $X \sim N(\mu, \sigma^2)$ is a normal random variable if the density is:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \ \forall -\infty < x < \infty$$

4 Expectation, Moments, and Counting Methods

A full distribution contains more information than many questions require. Expectations and moments summarize it, while linearity and indicators allow direct calculation without first finding the law.

4.1 Expectation, integrability, and variance

Definition 4.1. Expectation
For a discrete random variable, the expectation of X is defined by:

$$E(X) = \sum_{x \in \mathcal{S}_X} xP(X = x), \qquad E|X| < \infty$$

Definition 4.2. Expectation
For a continuous random variable X with $E|X| < \infty$, the expectation is

$$EX = E(X) = E[X] = \int_{-\infty}^{\infty} xf(x)\,dx$$

For any function g, we have:

$$E(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)\,dx$$

Note: Expectation is an integration of a measurement.

Theorem 4.1. Properties of expectation
Let X, Y be integrable random variables:

1. $\forall c \in \mathbb{R}, E(c) = c$
2. If $X \geq 0$, then $EX \geq 0$
3. If $c \in \mathbb{R}, E(cX) = cEX$
4. $E[X + Y] = E[X] + E[Y]$

By properties 3 and 4, we know that expectation is **linear**.

Proof sketch. Expectation is integration against a probability measure: constants integrate to themselves, non-negative functions have non-negative integrals, and the integral is linear for integrable functions. For example, if (X, Y) is discrete with joint mass $p(x, y)$, absolute convergence permits

$$E[X + Y] = \sum_{x,y} (x + y)p(x, y) = \sum_{x} xp_X(x) + \sum_{y} yp_Y(y).$$

The density case uses the same argument with integrals. No independence assumption is needed for linearity. □

For example: $X \sim Unif(0, 1)$

$$EX = \int_{-\infty}^{\infty} xf(x)\,dx = \frac{1}{2}$$

Definition 4.3. Variance
The variance of X is given by:

$$Var(X) = E[(X - EX)^2]$$

Moreover, we could also calculate by:

$$Var(X) = E(X^2) - (EX)^2$$

Quiz 4.1. Prove

$$\mu = \int_{-\infty}^{\infty} xf(x)\,dx \text{ and } \sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)\,dx$$

where:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

All finite expectation identities below assume the relevant variables are integrable. For a discrete variable, sum over its actual countable support, which need not consist of integers.

For signed variables, $E|X| < \infty$ ensures that positive and negative contributions do not produce an undefined difference.

Definition 4.4. Moments. For an integer $k \geq 1$, the raw moment is $E[X^k]$ and the central moment is $E[(X-\mu)^k]$, when defined. For $p > 0$, the absolute central moment is $E|X-\mu|^p$. The second central moment is the variance; odd central moments and odd absolute central moments are different quantities.

4.2 Moments of standard distributions

The following formulas use the parameter conventions stated in the preface.

Distribution	Mean	Variance
Bernoulli(p)	p	$p(1-p)$
Bin(n, p)	np	$np(1-p)$
Poisson(λ)	λ	λ
Geometric(p)	$1/p$	$(1-p)/p^2$
Unif(a, b)	$(a+b)/2$	$(b-a)^2/12$
Exp(λ)	$1/\lambda$	$1/\lambda^2$
Gamma(α, λ)	α/λ	α/λ^2
$\mathcal{N}(\mu, \sigma^2)$	μ	σ^2

For the Gamma law, integration gives $E[X^k] = \Gamma(\alpha+k)/(\lambda^k\Gamma(\alpha))$ for $k = 1, 2$. Using $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ yields the entries above.

4.3 Indicators, waiting times, and applications

Theorem 4.2. Properties of Expectation:
Let $X_1, X_2, \cdots, X_n$ be R.V.s such that $E|X_i| < \infty$, $\forall i$.

1. If $c_0, c_1, \cdots, c_n \in \mathbb{R}$, then:

$$E[c_0 + c_1X_1 + \cdots c_nX_n] = c_0 + c_1EX_1 + \cdots c_nEX_n$$

2. If $X_1, \ldots, X_n$ are independent and the measurable functions g_i satisfy $E|g_i(X_i)| < \infty$, then:

$$E[\prod_{k=1}^{n} g_k(X_k)] = \prod_{k=1}^{n} E[g_k(X_k)]$$

Remark: $g_1(X_1), \cdots, g_n(X_n)$ are also independent.

Proof sketch. The first assertion follows by repeated linearity. For the second, put $Z_i = g_i(X_i)$. Measurable functions of independent variables are independent, so their joint law is

the product of their marginal laws. Tonelli's theorem first gives

$$E\left[\prod_i |Z_i|\right] = \prod_i E|Z_i| < \infty.$$

Fubini's theorem then allows the signed product to be integrated one factor at a time, giving $E[\prod_i Z_i] = \prod_i E[Z_i]$. For discrete laws this is simply factorization of an absolutely convergent multiple sum. □

For example:
A group of n men and n women is lined up randomly.

1. Find the expected number of men with a woman beside them.

   Solution:

   Let:

$$X_i = \begin{cases} 1 & \text{, If man i has a woman next to him.} \\ 0 & \text{, Otherwise.} \end{cases}$$

$$EX_{total} = E[X_1 + \cdots + X_n] = \sum_{k=1}^{n} EX_k = nEX_1$$

$$EX_1 = P(X_1 = 1) = \frac{1}{2n} \cdot \frac{n}{2n-1} + \frac{1}{2n} \cdot \frac{n}{2n-1} + (2n-2) \cdot \frac{1}{2n} \cdot (1 - \frac{(n-1)(n-2)}{(2n-1)(2n-2)})$$

$$\Rightarrow EX_1 = \frac{3n-1}{4n-2}$$

$$\Rightarrow EX_{total} = \frac{n(3n-1)}{4n-2}$$

2. Repeat part 1°, assuming the group is randomly seated at a round table.

$$EX_{table} = n \times 2n \times \frac{1}{2n} \cdot (1 - \frac{(n-1)(n-2)}{(2n-1)(2n-2)}) = \frac{3n^2}{4n-2}$$

For example:
20 individuals, 10 married couples, five tables.

1. If the seating is random, find the expected number of couples seated at the same table.

$$EX_{total} = 10EX_1 = 10 \times \frac{3}{19} = \frac{30}{19}$$

2. If 2 men and 2 women are randomly chosen to be seated at each table, repeat 1°

$$EX_{total} = 10EX_1 = 10 \times \frac{2}{10} = 2$$

Quiz 4.2. Suppose there are n people coming to a party, they take turns to sit, for each table it has p probability not sitting it, if there is no table for him to sit, he will start a new table.

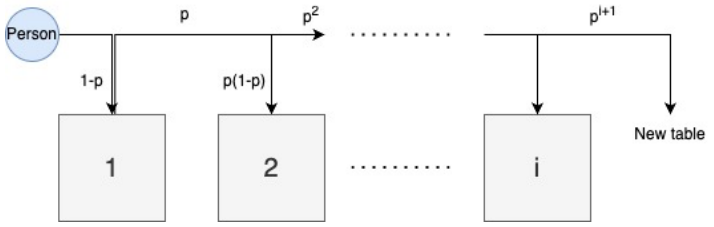


Figure 1: Sequential seating: each existing table is skipped with probability p.

What is the expectation of the numbers of the tables when all n people have taken their seats?

We now apply waiting-time decomposition to the coupon collector problem. (Coupon - Collection problem) Suppose there are N different types of coupons, and each time, it is equally likely to be any of the N types. Find the coupons needed before obtaining a complete set of all the N types.

Solution:

Define X_i, $0 \leq i \leq N-1$ as the number of additional coupons needed to obtain after i distinct types have been collected.

$$X_{total\ number} = X_0 + X_1 + \cdots + X_{N-1}$$

$$P(X_i = k) = \left(\frac{i}{N}\right)^{k-1} \cdot \frac{N-i}{N} \ \Rightarrow X_i \sim Geometric\left(\frac{N-i}{N}\right)$$

$$EX_{total\ number} = \sum_{i=0}^{N-1} EX_i = \sum_{i=0}^{N-1} E\left(Geometric\left(\frac{N-i}{N}\right)\right) = N \cdot \sum_{k=1}^{N} \frac{1}{k} \approx N \cdot lnN$$

4.4 Inclusion–exclusion through indicators

Theorem 4.3.

$$P(\bigcup_{i=1}^{n} E_i) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < i_2 < \cdots < i_k \leq n} P(\bigcap_{j=1}^{k} E_{i_j})$$

Proof. Write $I_i = \mathbf{1}_{E_i}$. Pointwise, $\prod_{i=1}^{n}(1 - I_i)$ equals one exactly when none of the events occurs. Consequently,

$$\mathbf{1}_{\cup_i E_i} = 1 - \prod_{i=1}^{n}(1 - I_i) = \sum_{\varnothing \neq J \subseteq \{1,\ldots,n\}} (-1)^{|J|+1} \prod_{j \in J} I_j.$$

Take expectations and use $\prod_{j \in J} I_j = \mathbf{1}_{\cap_{j \in J} E_j}$. Grouping the subsets J by their size gives the stated formula. Independence is not required: this is an identity of indicators before any probabilities are taken. □

4.5 Expectations from tail probabilities

Theorem 4.4. Tail formulas for non-negative variables. If $X \geq 0$, then

$$E[X] = \int_0^\infty P(X > t)\, dt.$$

If N is non-negative and integer-valued, then

$$E[N] = \sum_{k=1}^{\infty} P(N \geq k).$$

Both identities allow the value $+\infty$.

Proof. Use $X = \int_0^\infty \mathbf{1}_{\{X>t\}}\, dt$ or $N = \sum_{k\geq 1} \mathbf{1}_{\{N\geq k\}}$, then exchange expectation with a non-negative integral or sum. □

These formulas are particularly useful for waiting times: survival probabilities can be easier to compute than the full distribution.

5 Joint Distributions and Dependence

Joint distributions describe how quantities vary together. We first develop marginalization and independence, then use Gaussian models, covariance, and correlation to study dependence.

5.1 Joint laws, marginalization, and independence

Definition 5.1. Joint cumulative distribution function:
For any random variables X, Y the joint cumulative distribution function of X and Y is defined by:

$$F(a, b) = P(X \leq a, Y \leq b)$$

Obviously, by the axiom of probability, we should have:

$$\lim_{a,b\to\infty} (F(a, b)) = 1$$

Notice that:

$$P(X \leq a) = \lim_{b\to\infty} (P(X \leq a, Y \leq b))$$

Denote as $P(X \leq a) = F(a, \infty)$, and this is defined as **Marginal Cumulative Distribution Function**.

For discrete multi-random variables, we can define:

Definition 5.2. Joint probability mass function: When X, Y are both discrete random variables with p.m.f is given by p_X, p_Y.

The joint probability mass function:

$$p(X, Y) = P(X = x, Y = y)$$

Similar to the definition above, we have **Marginal Probability Mass Function:**

$$p_X(x) = \sum_y P(X = x, Y = y).$$

Similar to a single variable, we can also define independence in multi-variables.

Definition 5.3. Independent random variables
We say X, Y are independent if, for all Borel sets $A, B \subseteq \mathbb{R}$,

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

From the two definitions above, we could induce that:

Theorem 5.1. Two discrete random variables X, Y are independent if and only if $\forall x, y \in \mathbb{R}$,

$$p(x, y) = P_X(x)P_Y(y)$$

Proof sketch. Independence applied to singleton sets gives $p(x, y) = p_X(x)p_Y(y)$. Conversely, summing this factorization over the countable supports in any Borel sets A, B gives

$$P(X \in A, Y \in B) = \sum_{x \in A} \sum_{y \in B} p_X(x)p_Y(y) = P(X \in A)P(Y \in B),$$

which is the definition of independence. □

Definition 5.4. Jointly continuous
We say X and Y are jointly continuous if there exist a non-negative density $f(x, y)$ such that

$$\forall\ C \subset \mathbb{R}^2, P((X, Y) \in C) = \int_{(x,y)\in C} f(x, y)\, dxdy$$

The function $f(x, y)$ is called the **joint probability density function** of X and Y.

Definition 5.5. joint cumulative distribution function:
The joint c.d.f. is then given by:

$$F_{X,Y}(a, b) = \int_{-\infty}^{a} \int_{-\infty}^{b} f(x, y)\, dy\, dx$$

The definition of independence is still the same as before.

Definition 5.6. Expectation
For any joint p.m.f $p(x, y)$ or joint p.d.f $f(x, y)$, we have a $\forall$ function $g : \mathbb{R} \times \mathbb{R} \longrightarrow \mathbb{R}$,

$$E(g(X, Y)) = \sum_m \sum_n (g(m, n)p(m, n))$$

For example:

$$g(X, Y) = 1_{X \in A} 1_{Y \in B}$$

$$E[g(X, Y)] = E[1_{X \in A, Y \in B}] = P(X \in A, Y \in B) = \sum_{m \in A} \sum_{n \in B} p(m, n)$$

Or, in continuous situations:

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y)\, dxdy$$

We need to notice that $1_{X \in A}$ and $1_{Y \in B}$ are beneficial **Characteristic functions**.

Another Example: A man and a woman promised to meet at $12 : 30$P.M. Assume the time they arrive are X and Y independently and satisfy:

$$X \sim \text{Unif}(12 : 15, 12 : 45)$$

$$Y \sim \text{Unif}(12 : 00, 1 : 00)$$

1. Calculate P(the man arrive first)

$$X \sim \text{Unif}(-0.5, 0.5) Y \sim \text{Unif}(-1, 1)$$

$$\begin{aligned} P(X < Y) &= \int 1_{(X<Y)} \cdot f(x, y) \mathrm{d}x\, \mathrm{d}y \\ &= \frac{1}{2} \int_{-\frac{1}{2}}^{\frac{1}{2}} \mathrm{d}x \int_{x}^{1} \mathrm{d}y \\ &= \frac{1}{2} \int_{-\frac{1}{2}}^{\frac{1}{2}} (1 - x)\ \mathrm{d}x \\ &= \frac{1}{2} \end{aligned}$$

2. Find the probability that the first arrives waits no longer than 5 minutes.

$$\begin{aligned} P(|X - Y| < \frac{5}{30}) &= \iint 1_{(|X-Y|<\frac{1}{6})} \cdot f(x, y)\ \mathrm{d}x\, \mathrm{d}y \\ &= \frac{1}{2} \int_{-\frac{1}{2}}^{\frac{1}{2}} \mathrm{d}x \int_{x-\frac{1}{6}}^{x+\frac{1}{6}} \mathrm{d}y \\ &= \frac{1}{6} \end{aligned}$$

Quiz 5.1. The joint probability distribution function of X, Y is :

$$f(x,y) = \begin{cases} c & \text{if } x^2 + y^2 \leq R^2 \\ 0 & \text{otherwise} \end{cases}$$

a) Find c

b) Find the marginal probability distribution functions of $f_X(x)$ and $f_Y(y)$.

This is the uniform distribution in circle plates.

5.2 Gaussian models and covariance

Definition 5.7. Bivariate Normal Distribution
The joint probability of bivariate normal distribution is given by:

$$f(x,y) = \frac{\exp\left[-Q(x,y)/(2(1-\rho^2))\right]}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}},$$
$$Q(x,y) = \left(\frac{x-\mu_x}{\sigma_x}\right)^2 + \left(\frac{y-\mu_y}{\sigma_y}\right)^2 - 2\rho\frac{(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y}.$$

Here $\sigma_x, \sigma_y > 0$ and $|\rho| < 1$.
We denote it as $(X,Y) \sim N(\mu_x, \mu_y; \sigma_x^2, \sigma_y^2; \rho)$

Definition 5.8. Covariance
The covariance of X, Y is:

$$Cov(X,Y) := E[(X - EX)(Y - EY)]$$

By simple calculation, we know that:

$$Cov(X,X) = E[(X - EX)^2] = Var(X)$$

Now that we have expanded a single variable into bivariable, how can we get higher dimensions?

We could use **Matrix Form** to gain a beautiful expression of any finite dimension normal distribution.

If we let:

$$\begin{aligned}
\vec{x} &= (x, y) \\
\vec{\mu} &= (\mu_x, \mu_y) \\
\Sigma &= \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix} \\
\Rightarrow \quad \det(\Sigma) &= \sigma_x^2\sigma_y^2(1-\rho^2) \\
\Sigma^{-1} &= \frac{1}{1-\rho^2} \cdot \begin{bmatrix} \frac{1}{\sigma_x^2} & -\frac{\rho}{\sigma_x\sigma_y} \\ -\frac{\rho}{\sigma_x\sigma_y} & \frac{1}{\sigma_y^2} \end{bmatrix}
\end{aligned}$$

Then we have:

$$f(x,y) = \frac{1}{2\pi\sqrt{\det(\Sigma)}} \cdot \exp\left\{-\frac{1}{2}(\vec{x}-\vec{\mu})\Sigma^{-1}(\vec{x}-\vec{\mu})^T\right\}$$

This is an excellent structure with a general form.

Theorem 5.2. In Bivariate Normal Distribution, the X, Y independent are equivalent to:

1. $\rho = 0$
2. $Cov(X, Y) = 0$

Proof sketch. Let U, V be independent standard normals. The transformation

$$X = \mu_x + \sigma_x U, \qquad Y = \mu_y + \sigma_y\big(\rho U + \sqrt{1-\rho^2}\, V\big)$$

has the displayed bivariate normal density by a linear change of variables. It gives $\mathrm{Cov}(X, Y) = \rho\sigma_x\sigma_y$, so zero covariance is equivalent to $\rho = 0$. When $\rho = 0$, the joint density factors into the two marginal normal densities, proving independence. Conversely, independence implies zero covariance because $E[XY] = E[X]E[Y]$.

The Gaussian assumption matters: for $U \sim \mathrm{Unif}(-1, 1)$, the variables U and U^2 have zero covariance by symmetry, but are dependent. For example, the events $\{|U| \le 1/2\}$ and $\{U^2 \le 1/4\}$ coincide and have probability $1/2$, not product probability $1/4$. □

5.3 Variance of sums and correlation

The variance of X is given by:

$$Var(X) = E[(X - EX)^2] = \begin{cases} \sum_{m \in \mathcal{S}_X} (m-\mu)^2 P(X = m) \\ \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx \end{cases}$$

Now we introduce some properties of variance:

Theorem 5.3. Properties of variance:

1. If $EX = \mu$, $\mathrm{Var}(X) < \infty$, then:
   (a) $\mathrm{Var}(a + bX) = b^2\mathrm{Var}(X)$, $\forall\, a, b \in \mathbb{R}$
   (b) $\mathrm{Var}(X) = E[(X-\mu)^2] < E[(X-c)^2]$, $\forall\, c \in \mathbb{R} \neq \mu$
2. Let $X_1, X_2, \ldots, X_n$ be random variables with $Var(X_i) < \infty$, then:
   (a) $\mathrm{Var}(\sum_{i=1}^n X_i) = \sum_{i=1}^n \sum_{j=1}^n [E(X_i X_j) - \mu_i \mu_j]$
   (b) If $X_1, X_2, \ldots, X_n$ are independent.

$$Var(\sum_{i=1}^{n} X_i) = \sum_{i=1}^{n} Var(X_i)$$

Proof. Write $\mu = E[X]$. Centering removes the constant a, so

$$\mathrm{Var}(a + bX) = E[b^2(X-\mu)^2] = b^2\,\mathrm{Var}(X).$$

Also, expanding around the mean yields

$$E[(X-c)^2] = E[(X-\mu)^2] + 2(\mu - c)E[X-\mu] + (\mu-c)^2 = \mathrm{Var}(X) + (\mu-c)^2.$$

Thus the mean uniquely minimizes mean squared distance among constants. For $\mu_i = E[X_i]$, expand the square of the centered sum:

$$\begin{aligned}\mathrm{Var}\left(\sum_i X_i\right) &= \sum_{i,j} E[(X_i-\mu_i)(X_j-\mu_j)] \\ &= \sum_i \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j).\end{aligned}$$

Independent variables have zero covariance, so the cross terms vanish. In fact, pairwise uncorrelatedness already suffices for variance additivity. □

Definition 5.9. Correlation:
Let X, Y have finite, strictly positive variances.

$$\rho = \frac{Cov(X,Y)}{\sqrt{Var(X)}\sqrt{Var(Y)}}$$

To bound the magnitude of correlation, we use Cauchy–Schwarz.

Theorem 5.4. Cauchy–Schwarz Theorem:
For all random variables X, Y with finite second moments:

$$|E(XY)| \leq E(|XY|) \leq \sqrt{EX^2 \times EY^2}$$

Proof. The inequality $2|XY| \leq X^2 + Y^2$ ensures integrability. The triangle inequality gives

$|E[XY]| \le E|XY|$. If $E[Y^2] = 0$, then $Y = 0$ almost surely and the result is immediate. Otherwise, for every real a,

$$0 \le E[(|X| - a|Y|)^2] = E[X^2] - 2aE|XY| + a^2E[Y^2].$$

Choose $a = E|XY|/E[Y^2]$. The resulting inequality is $(E|XY|)^2 \le E[X^2]E[Y^2]$. Applying it to $X - E[X]$ and $Y - E[Y]$ proves $|\rho| \le 1$ whenever both variances are positive. □

The correlation coefficient therefore always belongs to $[-1, 1]$.

Definition 5.10. Uncorrelated:
If $\rho = 0$, or $Cov(X, Y) = 0$, we say X and Y are uncorrelated.

Obviously, if X, Y are independent, they must be uncorrelated.

6 Transformations and Sums of Random Variables

To pass from a model to a derived quantity, transform its distribution or combine contributions. The cumulative-distribution method, Jacobians, and convolution each preserve probability in a different calculation.

6.1 One-dimensional transformations and quantiles

The most general starting point is $F_Y(y) = P(g(X) \le y)$. Find this probability from the support of X, then differentiate where appropriate.

Theorem 6.1. If X has a density on an interval and g is continuously differentiable and strictly monotone there, with $g'(x) \neq 0$, then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

for y in the image of the interval, and zero otherwise.

Proof sketch. If g is increasing, then $F_Y(y) = F_X(g^{-1}(y))$. The chain rule gives the stated density almost everywhere. If g is decreasing, continuity of the law of X gives $F_Y(y) = 1 - F_X(g^{-1}(y))$; the inverse derivative is negative, so the same formula holds with its absolute value. Outside the image of the support, the density is zero. □

For a non-monotone transformation, divide the support into monotone branches and add their density contributions.

For example, if $X \sim \mathcal{N}(\mu, \sigma^2)$ and $a \neq 0$, then $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$. If $a = 0$, the result is the constant b.

Quiz 6.1. If the pdf of X is:

$$f(x) = \frac{1}{\pi(1 + x^2)}, \quad -\infty < x < \infty$$

Show that $Y = \frac{1}{X}$ has the same pdf.

Definition 6.1. Quantile function. For a c.d.f. F and $0 < u < 1$, define

$$Q(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}.$$

This generalized inverse remains meaningful when F has jumps or flat intervals.

Theorem 6.2. Inverse-transform sampling. If $U \sim \text{Unif}(0, 1)$, then $Q(U)$ has c.d.f. F.

Proof. For every x, the events $\{Q(U) \leq x\}$ and $\{U \leq F(x)\}$ agree, so their probability is $F(x)$. □

For example, $-\log(1 - U)/\lambda$ is exponential with rate λ. This provides a direct route from uniform samples to other distributions.

6.2 Multivariate transformations and simulation

A change of coordinates preserves probability, but changes density because it stretches or compresses area.

Theorem 6.3. Change of variables in two dimensions. Let $T : (x, y) \mapsto (g(x, y), h(x, y))$ be a continuously differentiable bijection between open regions, with a continuously differentiable inverse $(x, y) = (q(u, v), r(u, v))$ and nonzero Jacobian. Then

$$f_{U,V}(u, v) = f_{X,Y}(q(u, v), r(u, v)) \left| \det \frac{\partial(q, r)}{\partial(u, v)} \right|$$

on the transformed support, and zero elsewhere. If several inverse branches occur, sum the contributions of all branches.

Proof. For a measurable set B in the transformed region, apply the ordinary change-of-variables theorem for integrals:

$$\begin{aligned} P((U, V) \in B) &= \iint_{T^{-1}(B)} f_{X,Y}(x, y)\, dx\, dy \\ &= \iint_B f_{X,Y}(q(u, v), r(u, v)) \left| \det DT^{-1}(u, v) \right| \, du\, dv. \end{aligned}$$

The integrand is therefore the density. For finitely many inverse branches, partition the original region into one-to-one pieces (up to null sets) and add their probabilities. The absolute Jacobian measures area distortion, so its sign cannot affect probability. □

For independent standard normals, use $x = r\cos\theta$, $y = r\sin\theta$. Apart from a set of probability zero, choose $r > 0$ and $0 \leq \theta < 2\pi$. The absolute Jacobian is r, so

$$f_{R,\Theta}(r, \theta) = \frac{r}{2\pi} e^{-r^2/2}, \qquad r > 0, \quad 0 \leq \theta < 2\pi.$$

The density factors: R and Θ are independent, with $f_R(r) = re^{-r^2/2}$ for $r > 0$ and $\Theta \sim \mathrm{Unif}(0, 2\pi)$.

Definition 6.2. Box–Muller construction. For independent $U_1, U_2 \sim \mathrm{Unif}(0, 1)$, set

$$R = \sqrt{-2 \log U_1}, \qquad \Theta = 2\pi U_2, \qquad X = R \cos \Theta, \quad Y = R \sin \Theta.$$

Then X, Y are independent standard normal variables. Indeed, $F_R(r) = 1 - e^{-r^2/2}$ for $r \geq 0$, so the construction combines inverse-transform sampling with the polar-coordinate density above.

6.3 Gaussian moments from polar coordinates

For a standard normal Z, symmetry and the substitution $t = z^2/2$ give

$$E|Z|^\alpha = \frac{2^{\alpha/2}\Gamma((\alpha+1)/2)}{\sqrt{\pi}}, \qquad \alpha > -1.$$

For $\alpha \leq -1$, the integral diverges at zero. This illustrates why the existence of a moment must be checked before applying a formula.

For example: If $X, Y \sim \mathcal{N}(0, 1)$, independent, find $E(|X^2 + Y^2|^\alpha)$ for $\alpha \in \mathbb{R}$

Solution:

$$\begin{aligned}
E(|X^2 + Y^2|^\alpha) &= \iint (x^2 + y^2)^\alpha \cdot f_{XY}(x, y) dx dy \\
&= \iint (x^2 + y^2)^\alpha \cdot \frac{1}{2\pi} \cdot e^{-\frac{1}{2}(x^2+y^2)} dx dy \\
&= \int_0^\infty r dr \cdot \int_0^{2\pi} d\theta \cdot (r^2)^\alpha \cdot \frac{1}{2\pi} \cdot e^{-\frac{1}{2}r^2} \\
&= \int_0^\infty r^{2\alpha+1} \cdot e^{-\frac{1}{2}r^2} dr \\
&\overset{t=\frac{1}{2}r^2}{=\!=\!=} 2^\alpha \int_0^\infty t^\alpha e^{-t}\, dt \\
&= \begin{cases} 2^\alpha \cdot \Gamma(\alpha+1) & \forall \alpha > -1 \\ \infty & \forall \alpha \leq -1 \end{cases}
\end{aligned}$$

6.4 Convolution of independent variables

To understand the structure below better, we introduce the convolution of two functions.

Definition 6.3. Convolution: Let f and g be two functions

$$(f * g)(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau)\, d\tau$$

Theorem 6.4. Sum of independent random variables
Let X, Y be independent continuous random variables. And $Z = X + Y$, then we have:

$$F_Z(z) = \int_{-\infty}^{\infty} F_X(z-y) f_Y(y) dy$$

Differentiate to obtain:

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy \\ &= (f_X * f_Y)(z) \end{aligned}$$

Proof. Independence gives joint density $f_X(x)f_Y(y)$. Thus, by integrating over the half-plane $x + y \leq z$,

$$F_Z(z) = \int_{\mathbb{R}} \int_{-\infty}^{z-y} f_X(x) f_Y(y)\, dx\, dy = \int_{\mathbb{R}} F_X(z-y) f_Y(y)\, dy.$$

To justify the density without an informal interchange of differentiation and integration, set $(z, y) = (x + y, y)$. The inverse map $(x, y) = (z - y, y)$ has Jacobian determinant one. Its joint density is $f_X(z-y)f_Y(y)$, and integrating out y gives the convolution formula almost everywhere. □

Now we compute an example:
$X, Y \sim Exp(\lambda)$, and they are independent.
Compute the p.d.f. of $X + Y$:
Solution:

$$\begin{aligned} f_{X+Y}(z) &= \int_{-\infty}^{\infty} f_x(z-y) f_Y(y)\, dy \\ &= \int_0^z \lambda e^{-\lambda(z-y)} \times \lambda e^{-\lambda y}\, dy \\ &= \lambda^2 e^{-\lambda z} z \end{aligned}$$

By observation, we can easily notice that $X + Y \sim Gamma(2, \lambda)$

Quiz 6.2. If the joint p.d.f. of (X, Y) is :

$$f(x,y) = \begin{cases} \frac{1}{2}(x+y)e^{-(x+y)}, & 0 < x, y < \infty \\ 0 & \text{otherwise} \end{cases}$$

Find the p.d.f. of $Z = X + Y$.

6.5 Closure properties of discrete distributions

We begin with some easy conclusions:

Theorem 6.5. Here, we discuss the sum of independent discrete random variables.

1. **Poisson:** If $X \sim Poisson(\lambda_1)$ and $Y \sim Poisson(\lambda_2)$ are two independent discrete

random variables, then $X + Y \sim Poisson(\lambda_1 + \lambda_2)$

2. **Binomial:** If $X \sim Bin(n, p)$ and $Y \sim Bin(m, p)$ are two independent discrete random variables, then $X + Y \sim Bin(n + m, p)$

Proof. For independent Poisson variables and $k \geq 0$, discrete convolution and the binomial theorem give

$$\begin{aligned} P(X + Y = k) &= e^{-(\lambda_1+\lambda_2)} \sum_{j=0}^{k} \frac{\lambda_1^j \lambda_2^{k-j}}{j!(k-j)!} \\ &= e^{-(\lambda_1+\lambda_2)} \frac{(\lambda_1 + \lambda_2)^k}{k!}. \end{aligned}$$

For the binomial case, represent the two independent counts by disjoint collections of n and m independent Bernoulli(p) trials. Their sum counts successes in all $n + m$ trials and is therefore $\text{Bin}(n + m, p)$. Equivalently, discrete convolution uses Vandermonde's identity

$$\sum_{j=\max(0,k-m)}^{\min(n,k)} \binom{n}{j}\binom{m}{k-j} = \binom{n+m}{k},$$

which counts a k-element subset by how many elements it takes from the first group. □

7 Conditional Distributions and Conditional Expectation

Knowing part of a random system often makes the rest simpler. Conditional laws formalize this simplification; total expectation and total variance explain how to average the results back to the original model.

7.1 Conditional mass functions and densities

Definition 7.1. Conditional probability mass function:
If X, Y are two discrete random variables; we define the conditional probability mass function of X given $Y = y$:

$$\begin{aligned} P_{X|Y}(x|y) &= P(X = x|Y = y) \\ &= \frac{p(x, y)}{P_Y(y)} \end{aligned}$$

For example:
Suppose the $p(x, y)$ of (X, Y) is:

$$\begin{aligned} p(0, 0) = 0.4 \quad p(0, 1) = 0.2 \\ p(1, 0) = 0.1 \quad p(1, 1) = 0.3 \end{aligned}$$

Find the conditional distribution of X given $Y = 1$.

proof:

$$p(Y=1) = p(0,1) + p(1,1) = 0.5$$
$$P(X=0|\,Y=1) = \frac{P(X=0,Y=1)}{P(Y=1)} = \frac{p(0,1)}{0.5} = \frac{2}{5}$$
$$P(X=1|\,Y=1) = \frac{P(X=1,Y=1)}{P(Y=1)} = \frac{p(1,1)}{0.5} = \frac{3}{5}$$

Similarly, we can also define:

Definition 7.2. Conditional cumulative distribution function: If X, Y are two discrete random variables; we define the conditional cumulative distribution function of X given $Y = y$:

$$\begin{aligned} F_{X|Y}(x|y) &= P(X \le x | Y = y) \\ &= \sum_{m \le x} P(X = m | Y = y) \\ &= \sum_{m \le x} \frac{P(m, y)}{P(Y = y)} \end{aligned}$$

For example:

If X and Y are independent R.V.s with $X \sim Poisson(\lambda_1)$, $Y \sim Poisson(\lambda_2)$, calculate the conditional distribution of X given $X + Y = n$.

proof: $\forall\, 0 \le k \le n$:

$$\begin{aligned} P(X = k|\, X + Y = n) &= \frac{P(X = k, X + Y = n)}{P(X + Y = n)} \\ &= \frac{e^{-\lambda_1} \cdot \frac{\lambda_1^k}{k!} \cdot e^{-\lambda_2} \cdot \frac{\lambda_2^{n-k}}{(n-k)!}}{e^{-(\lambda_1+\lambda_2)} \cdot \frac{(\lambda_1+\lambda_2)^n}{n!}} \\ &= \frac{n!}{k!(n-k)!} \cdot \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k \cdot \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-k} \\ &= \binom{n}{k} \cdot \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k \cdot \left(1 - \frac{\lambda_1}{\lambda_1+\lambda_2}\right)^{n-k} \\ &\sim Bin(n, \frac{\lambda_1}{\lambda_1+\lambda_2}) \end{aligned}$$

Definition 7.3. Conditional density. If (X, Y) has joint density $f_{X,Y}$ and $f_Y(y) > 0$, define

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

This density integrates to one in x. It describes conditioning through densities, even though $P(Y = y) = 0$. Values at conditioning points with $f_Y(y) = 0$ are not determined by this ratio.

Quiz 7.1. Joint probability density function of (X, Y) is:

$$f(x,y) = \begin{cases} xe^{-x(y+1)} & 0 < x, y < \infty \\ 0 & \text{otherwise} \end{cases}$$

1. Find $f_{X|Y}(x|y)$
2. Find $f_{Y|X}(y|x)$

Now, we can see the **Total Probability Formula** from a new perspective.

Definition 7.4. Total Probability Formula:
Let $A_1, \ldots, A_n$ be mutually exclusive and $S = \bigcup_{k=1}^{n} A_k$. Then:

$$P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i)$$

The continuous version:

$$P(B) = \int_{-\infty}^{\infty} P(B|Y = y) f_Y(y)\, dy$$

7.2 Conditional Gaussian laws and support constraints

For a jointly normal pair with means μ_x, μ_y, positive standard deviations σ_x, σ_y, and $|\rho| < 1$, completing the square in $f(x, y)/f_Y(y)$ gives

$$X \mid (Y = y) \sim \mathcal{N}\left(\mu_x + \rho\frac{\sigma_x}{\sigma_y}(y - \mu_y), \sigma_x^2(1 - \rho^2)\right).$$

Conditioning shifts the mean linearly in the observation and reduces the variance. When $\rho = 0$, the conditional law equals the marginal law, consistent with independence.

Quiz 7.2. The joint pdf of (X, Y) is:

$$f(x,y) = \begin{cases} 2xe^{x^2-y} & 0 < x < 1, y > x^2 \\ 0 & \text{otherwise} \end{cases}$$

Find $f_{Y|X}(y|x)$ and $P(Y \geq \frac{1}{4}|X = x)$

See the solution sketch for the support-sensitive calculation.

7.3 Conditional expectation and recursive models

Recall the conditional probability density function of X given $Y = y$ is:

$$f_{X|Y} = \frac{f(x,y)}{f_Y(y)}$$

Now we can define:

Definition 7.5. Conditional expectation:
Let X and Y be two random variables:

$$E(X|Y=y) = \int x f_{X|Y}(x|y)\,dx = \int x \frac{f(x,y)}{f_Y(y)}\,dx$$

Theorem 7.1. Law of total expectation:
If (X,Y) has a joint density and $E|X| < \infty$, then:

$$E(X) = \int E(X|Y=y) f_Y(y)\,dy$$

Proof. Assume (X,Y) has a joint density and $E|X| < \infty$. On the set $f_Y(y) > 0$, substitute the definition of conditional density:

$$\begin{aligned}\int E[X \mid Y=y] f_Y(y)\,dy &= \iint x f_{X|Y}(x \mid y) f_Y(y)\,dx\,dy \\ &= \iint x f_{X,Y}(x,y)\,dx\,dy = E[X].\end{aligned}$$

Fubini's theorem is applicable because the integral with $|x|$ is $E|X| < \infty$. Values where $f_Y(y) = 0$ do not affect the integral. The same argument with sums proves the discrete version; in general the identity is $E[E[X \mid Y]] = E[X]$. □

For example:

A miner is trapped in a mine containing three doors. The first door leads to a tunnel that will take him to safety after 3 hours of travel. The second door leads to a tunnel that will return him to the mine after 5 hours of travel. The third door leads to a tunnel that will return him to the mine after 7 hours. If we assume that the miner is at all times equally likely to choose any one of the doors, what is the expected length of time until he reaches safety?

Solution:

Let X be the total time until safety and Y be the first choice of the door.

$$X = \begin{cases} 3 & \frac{1}{3} \\ 5 + X' & \frac{1}{3} \\ 7 + X' & \frac{1}{3} \end{cases}$$

Since the miner is equally likely to choose any of the doors at all times. So we know

$$\begin{cases} EX &= \frac{1}{3}E(X|Y=1) + \frac{1}{3}E(X|Y=2) + \frac{1}{3}E(X|Y=3) \\ EX &= EX' \end{cases}$$

So:

$$EX = \frac{1}{3}(3+5+7) + \frac{2}{3}EX$$

$$EX = 15$$

Quiz 7.3. Joint probability density function of X, Y is:

$$f(x,y) = \frac{1}{y} e^{-y-\frac{x}{y}}, x > 0, y > 0$$

Find $EX, EY, Cov(X,Y)$

7.4 Decomposing variance by conditioning

Theorem 7.2. Law of total variance. If $E[X^2] < \infty$, then

$$\mathrm{Var}(X) = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(E[X \mid Y]).$$

Proof. Put $m(Y) = E[X \mid Y]$ and $\mu = E[X]$. The centered variable decomposes as

$$X - \mu = (X - m(Y)) + (m(Y) - \mu).$$

Its two terms are orthogonal in expectation: conditioning on Y gives

$$\begin{aligned} E[(X - m(Y))(m(Y) - \mu)] &= E[(m(Y) - \mu)E[X - m(Y) \mid Y]] \\ &= 0. \end{aligned}$$

Here the conditional mean of the residual is zero, and $E[m(Y)] = \mu$ by total expectation. Squaring the decomposition and taking expectations now yields

$$\mathrm{Var}(X) = E[(X - m(Y))^2] + E[(m(Y) - \mu)^2] = E[\mathrm{Var}(X \mid Y)] + \mathrm{Var}(m(Y)).$$

All terms are finite: conditional Cauchy–Schwarz gives $m(Y)^2 \le E[X^2 \mid Y]$, and hence $E[m(Y)^2] \le E[X^2]$. □

The two terms represent variation remaining after Y is known and variation in the conditional mean. For the conditional normal model above they are $\sigma_x^2(1-\rho^2)$ and $\rho^2\sigma_x^2$, which sum to σ_x^2.

8 Order Statistics

Sorting observations changes the question from the identity of each sample point to its rank. The minimum, maximum, and sample quantiles are all order statistics.

8.1 Joint and marginal densities of ordered samples

Definition 8.1. Order Statistic:
Let $X_1, X_2 \ldots, X_n$ be independent identically distributed random variables with a common probability density function f and cumulative distribution function F

Then we define an ordered sequence of $X_1, X_2, \ldots, X_n$

$$X_{(1)} < X_{(2)} < \cdots < X_{(n)}$$

Theorem 8.1. The joint probability density function of $X_{(1)}, X_{(2)}, \ldots, X_{(n)}$ is:

$$f_{X_{(1)},X_{(2)},\ldots,X_{(n)}}(x_1, x_2, \ldots, x_n) = n! \times f(x_1)f(x_2)\ldots f(x_n) \times 1_{x_1<x_2<\cdots<x_n}$$

Proof. Ties have probability zero because the observations have densities. For any measurable set B contained in $x_1 < \cdots < x_n$, its inverse image under sorting consists of $n!$ disjoint coordinate permutations of B. The original joint density $\prod_i f(x_i)$ is unchanged by these permutations. Integrating over all $n!$ pieces therefore gives

$$P((X_{(1)}, \ldots, X_{(n)}) \in B) = n! \int_B \prod_{i=1}^{n} f(x_i)\, dx_1 \cdots dx_n,$$

which identifies the asserted density almost everywhere. □

For $1 \le j \le n$, the marginal density of the jth order statistic is

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!(n-j)!}[F(x)]^{j-1}[1-F(x)]^{n-j} f(x).$$

Derivation of the marginal density. Let $N_x = \sum_{i=1}^{n} \mathbf{1}_{\{X_i \le x\}}$. Independence gives $N_x \sim \mathrm{Bin}(n, F(x))$, and $\{X_{(j)} \le x\} = \{N_x \ge j\}$. Therefore

$$F_{X_{(j)}}(x) = \sum_{k=j}^{n} \binom{n}{k} F(x)^k [1-F(x)]^{n-k}.$$

Differentiate this finite sum at points where $F'(x) = f(x)$. Using $k\binom{n}{k} = n\binom{n-1}{k-1}$ and $(n-k)\binom{n}{k} = n\binom{n-1}{k}$ makes adjacent terms cancel, leaving $n\binom{n-1}{j-1}F(x)^{j-1}[1-F(x)]^{n-j}f(x)$, as claimed. □

For i.i.d. $\mathrm{Unif}(0,1)$ observations this becomes

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!(n-j)!} x^{j-1}(1-x)^{n-j}, \qquad 0 < x < 1,$$

and is zero otherwise. Thus $X_{(j)} \sim \mathrm{Beta}(j, n-j+1)$. A continuous variable satisfies $P(X = x) = 0$; a density value is not a point probability.

8.2 Extremes and the Beta family

For i.i.d. observations with continuous c.d.f. F, independence gives

$$P(X_{(n)} \le x) = F(x)^n, \qquad P(X_{(1)} > x) = [1-F(x)]^n.$$

Differentiating these expressions recovers the cases $j = n$ and $j = 1$ of the general order-statistic density. For a uniform sample, $X_{(j)} \sim \mathrm{Beta}(j, n-j+1)$ and hence $E[X_{(j)}] = j/(n+1)$.

9 Generating Functions and Characteristic Functions

Transforms encode probabilities or moments as coefficients and derivatives. They are especially effective for independent sums, because convolution becomes multiplication.

9.1 Probability generating functions

Definition 9.1. Probability generating function:
For a non-negative integer-valued random variable X, define :

$$|s| \leq 1, \qquad G_X(s) = E[s^X] = \sum_{n=0}^{\infty} s^n P(X = n)$$

is a function of s

The series is convergent at least for $|s| \leq 1$.
For example:

E.g.1 If $X \sim Bernoulli(p)$, then:

$$G_X(s) = 1 - p + sp$$

E.g.2 If $X \sim Poisson(\lambda)$, then:

$$\begin{aligned} G_X(s) &= \sum_{n=0}^{\infty} s^n e^{-\lambda} \frac{\lambda^n}{n!} \\ &= e^{-\lambda} \sum_{n=0}^{\infty} \frac{(s\lambda)^n}{n!} \\ &= e^{s\lambda - \lambda} \end{aligned}$$

E.g.3 If $X \sim Binomial(n, p)$, then:

$$G_X(s) = (1 - p + sp)^n$$

Theorem 9.1. Properties of probability generating function:

1. $$\forall\, k \geq 0, P(X = k) = \frac{1}{k!} G^{(k)}(0).$$

2. (a) $G_X(1) = 1$
   (b) $G^{(1)}(1) = EX$
   (c) $G^{(2)}(1) = E[X(X-1)]$
   (d) By the property above:

$$Var(X) = EX^2 - (EX)^2 = G^{(2)}(1) + G^{(1)}(1) - (G^{(1)}(1))^2$$

3. If $X_1, X_2, \ldots, X_n$ are independent, then for $Y = \sum_{i=1}^{n} X_i$, we have:

$$G_Y(s) = E[s^Y] = E[s^{X_1} \ldots s^{X_n}] = \prod_{i=1}^{n} G_{X_i}(s)$$

Proof. Inside $|s| < 1$, a power series can be differentiated term by term. At $s = 0$, its kth derivative isolates the coefficient $k!P(X = k)$. For $0 < s < 1$,

$$G'_X(s) = \sum_{k\geq 1} k s^{k-1} P(X = k), \qquad G''_X(s) = \sum_{k\geq 2} k(k-1) s^{k-2} P(X = k).$$

Let $s \uparrow 1$ and apply monotone convergence to obtain the first and second factorial moments, possibly infinite. Derivatives at 1 here mean limits from the left; the variance identity is used when $E[X^2] < \infty$. Finally, independence factors $E[\prod_i s^{X_i}]$ into a product of expectations, proving the generating-function rule for sums. □

9.2 Moment generating functions

Definition 9.2. Moment generating function:
If $M_X(t)$ is finite in a neighborhood of 0, its power series is:

$$M_X(t) = E(e^{tX}) = \sum_{n=0}^{\infty} \frac{t^n}{n!} E[X^n]$$

Remark: It is possible that $E(X^n) < \infty, \forall$ but $M_X(t) = \infty$

Quiz 9.1. The probability density function of X is;

$$f(x) = \frac{1}{2} e^{-|x|}, \forall\, x \in \mathbb{R}$$

1. Compute $M_X(t)$.
2. Compute $E[X^n]$ by $M_X(t)$.

9.3 Characteristic functions and inversion

Definition 9.3. Characteristic function:
The characteristic function of X is defined by :

$$\Phi_X(t) = E[e^{itX}] = E[\cos(tX)] + iE[\sin(tX)], \forall\, t \in \mathbb{R}$$

The cumulative distribution function of X is uniquely determined by $\Phi(t)$.

$$\Phi(t) = \int_{-\infty}^{\infty} e^{itx} dF(x)$$

For example:

$$\text{If} X \sim N(\mu, \sigma^2), \text{ then } E[e^{itX}] = e^{it\mu - \frac{1}{2}\sigma^2 t^2}$$

Inversion Formula:

$\forall\, a < b \in \mathbb{R}$.

$$\frac{1}{2\pi} \lim_{T\to\infty} \int_{-T}^{T} \frac{e^{-ita} - e^{-itb}}{it} \Phi(t)\, dt = \frac{1}{2}(F(b) + F(b^-)) - \frac{1}{2}(F(a) + F(a^-))$$

Equals to $F(b) - F(a)$ if left continuous.

10 Probability Inequalities

When an exact distribution is unavailable, moment information can still control tails. The inequalities below differ in the information they require and the events they bound.

10.1 Markov and Chebyshev inequalities

Theorem 10.1. Markov's Inequality:
For any random variable X with $E|X| < \infty$, we have:

$$P(|X| \geq t) \leq \frac{E|X|}{t}, \qquad t > 0.$$

Proof. The pointwise inequality $|X| \geq t\mathbf{1}_{\{|X|\geq t\}}$ gives

$$E|X| \geq tE[\mathbf{1}_{\{|X|\geq t\}}] = tP(|X| \geq t).$$

Divide by $t > 0$. This argument also shows the usual form $P(W \geq t) \leq E[W]/t$ for any non-negative W. □

Theorem 10.2. Chebyshev's Inequality:
For any random variable X with mean μ, variance σ^2, we have:

$$P(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}, \ \ \forall t > 0$$

Proof. Apply Markov's inequality to the non-negative variable $W = (X - \mu)^2$ with threshold t^2. Then

$$P(|X - \mu| \geq t) = P(W \geq t^2) \leq \frac{E[W]}{t^2} = \frac{\sigma^2}{t^2}.$$

□

Quiz 10.1. Suppose $X \geq 0$ has $EX = 10$.

1. Find a upper bound of $P(X > 15)$.
2. Suppose $Var(X) = 3$. Find a better bound for (a).

10.2 Chernoff bounds

Theorem 10.3. Chernoff Bound:
For any real threshold a:

- $P(X \geq a) \leq \inf_{t>0} e^{-ta} E[e^{tX}]$
- $P(X \leq a) \leq \inf_{t<0} e^{-ta} E[e^{tX}]$

Proof. For $t > 0$, the event $\{X \geq a\}$ equals $\{e^{tX} \geq e^{ta}\}$. Markov's inequality gives $P(X \geq a) \leq e^{-ta} E[e^{tX}]$. If $t < 0$, the exponential reverses the order, so the same argument bounds $P(X \leq a)$. Each bound holds for every admissible t; take the infimum. Parameters where the expectation is infinite give only a vacuous bound. □

Application of this Chernoff bound:
For $Z \sim N(0, 1)$,

$$P(Z \geq a) \leq \frac{E[e^{tZ}]}{e^{ta}} = e^{\frac{1}{2}t^2 - at}$$

Since this is correct for all $t > 0$.

$$P(Z \geq a) \leq e^{-\frac{1}{2}a^2} \quad \forall\, a > 0$$

This is a pretty good estimation for the normal distribution.

10.3 Convexity and Jensen's inequality

Theorem 10.4. Jensen's Inequality:
Let f be convex on an interval containing the values of an integrable random variable X. If $f(X)$ is integrable, then:

$$f(E[X]) \leq E[f(X)]$$

Proof. Let $m = E[X]$. If m is in the interior of the interval, convexity provides a supporting line at m: there exists a real c such that

$$f(x) \geq f(m) + c(x - m)$$

throughout the interval. (For differentiable f, take $c = f'(m)$; otherwise choose c between the one-sided derivatives.) Taking expectations eliminates the linear term and gives $E[f(X)] \geq f(m)$.

If m is an endpoint of the interval, the one-sided random variable $X - m$ has mean zero and therefore equals zero almost surely. In that case $X = m$ almost surely and equality holds. This proof explains Jensen's inequality geometrically: a convex graph lies above any of its supporting lines. □

11 Convergence and Limit Theorems

Limit theorems explain how random behavior stabilizes as sample size grows. We distinguish modes of convergence before studying averages, fluctuations, and conditions that justify passing limits through expectations.

11.1 Modes of convergence

Definition 11.1. Four types of convergence:

1. **convergence in probability:**(p)
   Let $X_1, X_2, \ldots, X_n$ be a sequence of random variables. We say X_n converges to X in probability if:

$$\forall\, \epsilon > 0,\ P(|X_n - X| > \epsilon) \to 0 (n \to \infty)$$

2. **almost surely convergence:**(a.s.)

$$P(\lim_{n\to\infty} X_n = X) = 1$$

3. **Convergence in distribution:**(d)
   For any $x \in \mathbb{R}$ such that x is a continuity point of $F_X(b) = P(X \leq b)$

$$\lim_{n\to\infty} P(X_n \leq x) = P(X \leq x)$$

4. L^p **convergence for** $p > 0 (L^p)$

$$\lim_{n\to\infty} E|X_n - X|^p = 0$$

From strong convergence to weak convergence:

$$L^p \to p \to d$$
$$a.s. \to p \to d$$

In general, neither L^p convergence nor almost sure convergence implies the other without additional assumptions.

Why the convergence implications hold. For L^p convergence, Markov's inequality gives

$$P(|X_n - X| > \epsilon) \leq \epsilon^{-p} E|X_n - X|^p \longrightarrow 0.$$

For almost sure convergence, fix $\epsilon > 0$ and define $B_N = \bigcup_{n \geq N} \{|X_n - X| > \epsilon\}$. Then $B_N \downarrow B$ with $P(B) = 0$, since a convergent sequence eventually stays within ϵ of its limit. For $n \geq N$, the deviation probability is at most $P(B_N) \to 0$.

Finally, if $X_n \to X$ in probability, then for any $\delta > 0$,

$$F_X(x-\delta) - P(|X_n - X| > \delta) \le F_{X_n}(x) \le F_X(x+\delta) + P(|X_n - X| > \delta).$$

First let $n \to \infty$, then $\delta \downarrow 0$. At every continuity point of F_X, the two bounds converge to $F_X(x)$, proving convergence in distribution. □

11.2 Weak and strong laws of large numbers

Chebyshev's inequality turns a variance calculation into a quantitative law of large numbers.

Theorem 11.1. Weak Law of Large Numbers:
Let $(X_n)_{n\ge1}$ be i.i.d. random variables with mean μ and finite variance σ^2. Then:

$$\forall\, \epsilon > 0,\ \lim_{n\to\infty} P\left(\left|\frac{X_1 + X_2 + \cdots + X_n}{n} - \mu\right| > \epsilon\right) = 0.$$

Proof. For $\overline{X}_n = n^{-1}\sum_{i=1}^n X_i$, linearity gives $E[\overline{X}_n] = \mu$ and independence gives

$$\mathrm{Var}(\overline{X}_n) = \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}(X_i) = \frac{\sigma^2}{n}.$$

For every fixed $\epsilon > 0$, Chebyshev's inequality now yields

$$P(|\overline{X}_n - \mu| > \epsilon) \le \frac{\sigma^2}{n\epsilon^2} \longrightarrow 0.$$

The bound controls deviation probabilities, which is precisely convergence in probability. □

Theorem 11.2. Strong Law of large numbers:
Let $(X_n)_{n\ge1}$ be i.i.d. random variables with $E|X_1| < \infty$ and $E[X_1] = \mu$. Then

$$\frac{1}{n}\sum_{i=1}^n X_i \xrightarrow{a.s.} \mu.$$

Proof sketch. The full integrable case requires more than the Chebyshev argument. We outline the truncation proof, using the following standard auxiliary result without proving it here: if independent centered variables Z_n satisfy $\sum_n E[Z_n^2]/n^2 < \infty$, then $n^{-1}\sum_{k=1}^n Z_k \to 0$ almost surely. This follows from Kolmogorov's convergence theorem for $\sum_n Z_n/n$ and Kronecker's lemma.

Set $Y_n = X_n \mathbf{1}_{\{|X_n|\le n\}}$. Integrability and the tail formula give

$$\sum_{n\ge1} P(X_n \ne Y_n) = \sum_{n\ge1} P(|X_1| > n) \le E|X_1| < \infty.$$

By the Borel–Cantelli lemma (Theorem 11.5), only finitely many terms are changed, almost

surely. Moreover, Tonelli's theorem gives

$$\sum_{n\geq 1}\frac{\mathrm{Var}(Y_n)}{n^2}\leq E\left[|X_1|^2\sum_{n\geq\max(1,\lceil|X_1|\rceil)}\frac{1}{n^2}\right]\leq 2E|X_1|<\infty.$$

For $x\geq 1$, the series starting at $\lceil x\rceil$ is at most $2/x$; for $0\leq x<1$, use $\sum n^{-2}<2$ and $x^2\leq x$.

Apply the auxiliary result to $Z_n=Y_n-E[Y_n]$. Also $E[Y_n]=E[X_1\mathbf{1}_{\{|X_1|\leq n\}}]\to\mu$ by domination, so their arithmetic averages converge to μ. Thus the averages of Y_n converge almost surely to μ, and changing finitely many terms does not change the limit for X_n. □

11.3 The central limit theorem and transform methods

Theorem 11.3. Central Limit Theorem:
Let $X_1, X_2, \ldots, X_n$ be a sequence of i.i.d random variables, with mean μ and variance $0<\sigma^2<\infty$. Then:

$$\lim_{n\to\infty}P(\frac{\sum_{i=1}^n X_i-n\mu}{\sqrt{n\sigma^2}}\leq x)=P(Z\leq x),\forall\, x\in\mathbb{R}$$

Where $Z\sim N(0,1)$. That is:

$$\frac{\sum_{i=1}^n X_i-n\mu}{\sqrt{n\sigma^2}}\xrightarrow{d}Z\ (n\to\infty)$$

Proof. Standardize by $Y_i=(X_i-\mu)/\sigma$, so that $E[Y_i]=0$ and $E[Y_i^2]=1$. Write $\phi(u)=E[e^{iuY_1}]$. The second-order expansion is

$$\phi(u)=1-\frac{u^2}{2}+o(u^2)\qquad(u\to 0).$$

To justify it using only a second moment, set $r(v)=e^{iv}-1-iv+v^2/2$. Taylor's formula gives $r(v)/v^2\to 0$ as $v\to 0$, and $|r(v)|\leq v^2$ for all real v. Hence $r(uY_1)/u^2\to 0$ pointwise and is bounded in absolute value by Y_1^2; dominated convergence proves the expansion. No third moment or moment generating function is required.

Independence now turns the characteristic function of the normalized sum into

$$E\left[e^{itn^{-1/2}\sum_{i=1}^n Y_i}\right]=\phi(t/\sqrt{n})^n=\left(1-\frac{t^2}{2n}+o(n^{-1})\right)^n\longrightarrow e^{-t^2/2}.$$

The limit is the characteristic function of a standard normal variable. The continuity theorem (Theorem 11.4) below converts this pointwise convergence into convergence in distribution. Since the normal c.d.f. is continuous everywhere, the asserted c.d.f. limit holds for every real x. □

This describes the limiting distribution of the centered sum scaled by $\sqrt{n}$.

Theorem 11.4. Continuity Theorem:
Let $\xi_1, \xi_2, \ldots, \xi_n$ be a sequence of random variables with characteristic function $\Phi_n(t) = E[e^{it\xi_n}]$. If $\Phi_n(t) \to \Phi(t), \forall\, t \in \mathbb{R}$ and $\Phi(t) = E[e^{itX}]$ for some random variables X and $\Phi(t)$ is continuous at 0. Then:

$$\xi_n \xrightarrow{d} X$$

Proof sketch. The two ingredients are tightness and uniqueness of characteristic functions. Here tightness means that, for each $\epsilon > 0$, a bounded interval contains probability at least $1 - \epsilon$ for every member of the sequence. For $\delta > 0$, averaging the real part of a characteristic function gives

$$\begin{aligned} A_n(\delta) &= \frac{1}{2\delta} \int_{-\delta}^{\delta} (1 - \operatorname{Re} \Phi_n(t))\, dt \\ &= E\left[1 - \frac{\sin(\delta \xi_n)}{\delta \xi_n}\right] \geq \tfrac{1}{2} P(|\xi_n| \geq 2/\delta), \end{aligned}$$

with $\sin(0)/0 = 1$. For fixed δ, bounded convergence under the integral gives $A_n(\delta) \to A(\delta)$, the same expression with Φ. Continuity at zero and $\Phi(0) = 1$ imply $A(\delta) \to 0$ as $\delta \downarrow 0$. This controls the tails for all sufficiently large n; the finitely many remaining distributions can also be covered by a larger interval.

A standard compactness fact for probability laws on $\mathbb{R}$ says that every tight sequence has a subsequence converging in distribution. Any such subsequential limit has characteristic function Φ, since e^{itx} is bounded and continuous. Uniqueness of characteristic functions identifies its law with that of X. Every subsequence therefore has a further subsequence with the same limit law, forcing the whole sequence to converge. The compactness fact and the bounded-continuous-function characterization of convergence in distribution are used here without proof. □

Using **characteristic function sequence's** convergence, we can get convergence in distribution.

Quiz 11.1. Let $X_1, X_2, \ldots, X_{10}$ be i.i.d. $\sim Unif(0, 1)$.Let $S_{10} = \sum_{i=1}^{10} X_i$. Find $P(S_{10} > 6)$ using:

1. Markov's Inequality
2. Chebyshev's Inequality
3. Central limit theorem

From the quiz result, we can observe which is the most accurate.

11.4 Almost sure events and dominated convergence

Theorem 11.5. Borel-Cantelli Lemma:
For a sequence of events $\{A_n\}$ if $\sum_{k=1}^{\infty} P(A_k) < \infty$ then :

$$P(A_n \text{ infinitely often }) = 0$$

Proof. The event that infinitely many A_n occur is

$$\{A_n \text{ infinitely often}\} = \bigcap_{N=1}^{\infty} \bigcup_{n \geq N} A_n.$$

For each N, the union bound gives

$$P(A_n \text{ infinitely often}) \leq P\left(\bigcup_{n \geq N} A_n\right) \leq \sum_{n \geq N} P(A_n).$$

The tail of the convergent series tends to zero. Independence is not needed for this direction of the Borel–Cantelli lemma. □

Quiz 11.2. If $X_n \xrightarrow{p} X$ and $P(|\ Y\ | < \infty) = 1$, then prove that:

$$X_n Y \xrightarrow{p} XY$$

Theorem 11.6. Dominated Convergence:
Let $p > 0$. If $X_n \xrightarrow{a.s.} X$ and $|X_n|^p \leq Y$ for every n, where $Y \geq 0$ and $EY < \infty$, then:

$$E[|\ X_n - X\ |^p] \to 0$$

Proof. Work on the probability-one event where $X_n \to X$ and all domination inequalities hold. Passing to the limit gives $|X|^p \leq Y$. The elementary inequality

$$|a - b|^p \leq 2^{\max\{p-1,0\}}(|a|^p + |b|^p)$$

implies $0 \leq Z_n := |X_n - X|^p \leq C_p Y$, where $C_p = 2^{\max\{p,1\}}$, and $Z_n \to 0$ almost surely.

For completeness, the required integral limit follows from Fatou's lemma: for non-negative W_n, $E[\liminf W_n] \leq \liminf E[W_n]$. Apply it to $W_n = C_p Y - Z_n \geq 0$ to get

$$C_p E[Y] \leq \liminf_n (C_p E[Y] - E[Z_n]) = C_p E[Y] - \limsup_n E[Z_n].$$

Thus $\limsup_n E[Z_n] \leq 0$; non-negativity gives $E[Z_n] \to 0$. Fatou's lemma is the measure-theoretic input in this argument. □

Domination supplies the extra control needed to upgrade almost sure convergence to L^p convergence.

A Answers and Solution Sketches

The sketches below highlight the main idea and the essential calculation. Each heading refers to the corresponding quiz in the lecture notes.

Quiz 1.1: Counting committees

Idea. Count all committees, then subtract those containing the forbidden pair. There are $\binom{6}{3}\binom{8}{3} = 1120$ unrestricted committees.

1. Two specified men cannot both serve:

$$\left[\binom{6}{3} - \binom{4}{1}\right]\binom{8}{3} = 896.$$

2. Two specified women cannot both serve:

$$\binom{6}{3}\left[\binom{8}{3} - \binom{6}{1}\right] = 1000.$$

3. A specified man and woman cannot both serve:

$$1120 - \binom{5}{2}\binom{7}{2} = 910.$$

Quiz 1.2: Increasing sequences

Idea. Every ordered sample has probability N^{-n}. A strictly increasing sample is uniquely determined by its n distinct values, whereas a non-decreasing sample is determined by the multiplicities of its values. Thus

$$P(\text{strictly increasing}) = \frac{\binom{N}{n}}{N^n}, \qquad P(\text{non-decreasing}) = \frac{\binom{N+n-1}{n}}{N^n}.$$

The first probability is zero when $n > N$. For the second count, use stars and bars on $c_1 + \cdots + c_N = n$, where c_j counts occurrences of card j.

Quiz 2.1: Identifying the coin

Idea. Apply Bayes' theorem, keeping the chosen coin fixed across tosses. Let F denote the fair coin and H^r denote r consecutive heads. With equal prior probabilities,

$$P(F \mid H^r) = \frac{(1/2)2^{-r}}{(1/2)2^{-r} + (1/2)} = \frac{1}{1+2^r}.$$

For one and two heads the answers are $1/3$ and $1/5$. A subsequent tail is impossible for the two-headed coin, so the posterior probability of the fair coin becomes 1.

Quiz 3.1: Reading a mixed c.d.f.

Idea. Strict inequalities use left limits, and point masses are jumps of the c.d.f. Therefore

$$\begin{aligned} P(X < 3) &= F(3^-) = \frac{11}{12}, \\ P(X = 1) &= F(1) - F(1^-) = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}, \\ P(X > 1/2) &= 1 - F(1/2) = \frac{3}{4}. \end{aligned}$$

Quiz 3.2: The random walk

Idea. Count rightward steps. If $R_n \sim \mathrm{Bin}(n, p)$ is their number, then $X_n = R_n - (n - R_n) = 2R_n - n$. Hence, for an integer k,

$$P(X_n = k) = \begin{cases} \binom{n}{(n+k)/2} p^{(n+k)/2}(1-p)^{(n-k)/2}, & |k| \le n,\ n+k \text{ even}, \\ 0, & \text{otherwise.} \end{cases}$$

The parity condition expresses the fact that each step changes the position by one.

Quiz 4.1: Moments of a normal variable

Idea. Standardize with $z = (x - \mu)/\sigma$, where $\sigma > 0$, and write $\phi(z) = (2\pi)^{-1/2} e^{-z^2/2}$. Symmetry gives $\int z\phi(z)\,dz = 0$. Since $\phi'(z) = -z\phi(z)$, integration by parts gives

$$\int_{-\infty}^{\infty} z^2 \phi(z)\,dz = -[z\phi(z)]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \phi(z)\,dz = 1.$$

Substituting $x = \mu + \sigma z$ therefore yields $E[X] = \mu$ and $E[(X - \mu)^2] = \sigma^2$.

Quiz 4.2: Sequential seating

Idea. Track the number of occupied tables, rather than individual seating arrangements. Under the stated rule, a person independently skips each existing table with probability p, and opens a new table only after skipping them all. Thus, if $T_m = k$, the probability of a new table is p^k.

Let $q_{m,k} = P(T_m = k)$, with $q_{0,0} = 1$ and probabilities outside $0 \le k \le m$ equal to zero. For $1 \le k \le m + 1$,

$$q_{m+1,k} = (1 - p^k) q_{m,k} + p^{k-1} q_{m,k-1}.$$

This finite recursion determines the exact answer

$$E[T_n] = \sum_{k=1}^{n} k q_{n,k}.$$

Equivalently, $E[T_{m+1}] = E[T_m] + E[p^{T_m}]$. For example, $E[T_1] = 1$, $E[T_2] = 1 + p$, and $E[T_3] = 1 + 2p - p^2 + p^3$. In general, $E[p^{T_m}]$ cannot be replaced by $p^{E[T_m]}$.

Diagram note. With k existing tables the new-table probability is p^k; the final exponent in the supplied sketch should be interpreted accordingly. At the endpoints, $p = 0$ gives one

table and $p = 1$ gives n tables, for $n \geq 1$.

Quiz 5.1: Uniform distribution on a disk

Idea. Normalize by the area, then integrate along a vertical or horizontal chord. For $R > 0$, $c\pi R^2 = 1$, so $c = 1/(\pi R^2)$. For $|x| < R$,

$$f_X(x) = \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} c\, dy = \frac{2\sqrt{R^2 - x^2}}{\pi R^2}.$$

It is zero outside this interval. By symmetry, $f_Y(y) = 2\sqrt{R^2 - y^2}/(\pi R^2)$ for $|y| < R$, and zero otherwise.

Quiz 6.1: Reciprocal of a Cauchy variable

Idea. Apply the change-of-variable formula separately on the positive and negative half-lines. For $y \neq 0$, the inverse is $x = 1/y$ and the absolute derivative is $1/y^2$, so

$$f_Y(y) = f_X(1/y)\frac{1}{y^2} = \frac{1}{\pi(1 + 1/y^2)y^2} = \frac{1}{\pi(1 + y^2)}.$$

Since $P(X = 0) = 0$, the reciprocal is well-defined almost surely; the chosen value of a density at $y = 0$ does not affect the distribution.

Quiz 6.2: A sum from a joint density

Idea. Integrate the joint density along $x + y = z$. Independence is not needed. The transformation $(x, y) \mapsto (x, z = x + y)$ has absolute Jacobian 1. For $z > 0$,

$$f_Z(z) = \int_0^z f(x, z - x)\, dx = \int_0^z \frac{1}{2} z e^{-z}\, dx = \frac{1}{2} z^2 e^{-z}.$$

The density is zero for $z \leq 0$.

Quiz 7.1: Conditional densities

Idea. First integrate out one variable to find the marginals, then divide the joint density by the appropriate marginal:

$$f_X(x) = e^{-x} \quad (x > 0), \qquad f_Y(y) = \frac{1}{(y+1)^2} \quad (y > 0).$$

Consequently,

$$f_{X|Y}(x \mid y) = (y+1)^2 x e^{-x(y+1)}, \qquad f_{Y|X}(y \mid x) = x e^{-xy}, \qquad x, y > 0.$$

Each conditional density is zero outside its positive support.

Quiz 7.2: Conditioning on a curved support

Idea. Preserve the support $y > x^2$ when integrating. For $0 < x < 1$,

$$f_X(x) = \int_{x^2}^{\infty} 2xe^{x^2-y}\,dy = 2x, \qquad f_{Y|X}(y \mid x) = e^{x^2-y}\mathbf{1}_{\{y>x^2\}}.$$

Integrating from $\max\{x^2, 1/4\}$ gives

$$P(Y \geq 1/4 \mid X = x) = \begin{cases} e^{x^2-1/4}, & 0 < x < 1/2, \\ 1, & 1/2 \leq x < 1. \end{cases}$$

Quiz 7.3: Covariance by conditioning

Idea. Recognize $Y \sim \text{Exp}(1)$ and, given $Y = y$, X is exponential with mean y. Thus $E[X \mid Y] = Y$ and

$$E[Y] = 1, \qquad E[X] = E[Y] = 1, \qquad E[XY] = E\{YE[X \mid Y]\} = E[Y^2] = 2.$$

It follows that $\text{Cov}(X, Y) = 2 - 1 \cdot 1 = 1$.

Quiz 9.1: The Laplace moment generating function

Idea. Split the integral at zero:

$$\begin{aligned} M_X(t) &= \frac{1}{2}\int_0^{\infty} e^{-(1-t)x}\,dx + \frac{1}{2}\int_{-\infty}^{0} e^{(1+t)x}\,dx \\ &= \frac{1}{2(1-t)} + \frac{1}{2(1+t)} = \frac{1}{1-t^2}, \qquad |t| < 1. \end{aligned}$$

The integral diverges for $|t| \geq 1$. Comparing $M_X(t) = \sum_{k\geq 0} t^{2k}$ with $\sum_{n\geq 0} E[X^n]t^n/n!$ gives $E[X^{2k}] = (2k)!$ and $E[X^{2k+1}] = 0$.

Quiz 10.1: Tail bounds from moments

Idea. Use the information supplied at each stage. Markov's inequality gives

$$P(X > 15) \leq \frac{E[X]}{15} = \frac{2}{3}.$$

With $\text{Var}(X) = 3$, the event $X > 15$ implies $|X - 10| > 5$. Chebyshev's inequality therefore improves the bound to $3/25$.

Quiz 11.1: Bounds and a normal approximation

Idea. Compute the mean and variance of the sum first:

$$E[S_{10}] = 5, \qquad \text{Var}(S_{10}) = \frac{10}{12} = \frac{5}{6}.$$

Markov gives $P(S_{10} > 6) \leq 5/6$. Direct Chebyshev also gives $5/6$. Since $S_{10} - 5$ is symmetric and continuous, the two tails have equal probability, yielding the sharper Chebyshev-based

bound

$$P(S_{10} > 6) = \frac{1}{2}P(|S_{10} - 5| > 1) \leq \frac{5}{12}.$$

The central limit theorem suggests

$$P(S_{10} > 6) \approx 1 - \Phi\left(\frac{1}{\sqrt{5/6}}\right) = 1 - \Phi(\sqrt{6/5}) \approx 0.137,$$

where Φ is the standard normal c.d.f. This last expression is an approximation, not a guaranteed upper bound.

Quiz 11.2: Multiplication by a finite random variable

Idea. Truncate Y to a bounded event. For any $M > 0$ and $\varepsilon > 0$,

$$P(|(X_n - X)Y| > \varepsilon) \leq P(|Y| > M) + P(|X_n - X| > \varepsilon/M).$$

For fixed M, the second term tends to zero. Then let $M \to \infty$; the first term tends to zero because Y is finite almost surely. This proves $X_n Y \xrightarrow{p} XY$, without requiring independence.